LAB EXPERIMENT RECORD

Set 26: Dataset: Student_Mini_Data.csv

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
Date	20/10/2026
Day	Tuesday

AIM

To analyze student academic performance using R and Tableau by creating histograms, boxplots, scatter plots, monthly trend charts, and an interactive dashboard.

1.

```
student <- data.frame(
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
  Attendance = c(78, 90, 70, 95, 85, 92),
  Math_Score = c(62, 80, 55, 90, 72, 82),
  Science_Score = c(65, 85, 58, 92, 74, 86),
  Exam_Date = as.Date(c("2025-01-10", "2025-01-10",
                        "2025-02-12", "2025-02-12",
                        "2025-03-15", "2025-03-15"))
)

hist(student$Math_Score,
     main = "Distribution of Math Scores",
     xlab = "Math Score",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")

boxplot(Science_Score ~ Gender, data = student,
        main = "Science Scores by Gender",
        xlab = "Gender",
        ylab = "Science Score",
        col = c("pink", "lightgreen"))
```

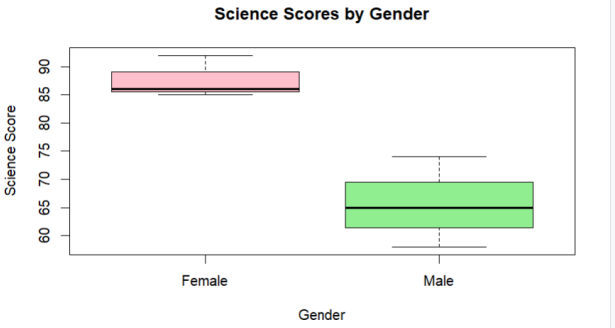
```
  Math_Score = c(62, 80, 55, 90, 72, 82),
  Science_Score = c(65, 85, 58, 92, 74, 86),
  Exam_Date = as.Date(c("2025-01-10", "2025-01-10",
                        "2025-02-12", "2025-02-12",
                        "2025-03-15", "2025-03-15"))
)

hist(student$Math_Score,
     main = "Distribution of Math Scores",
     xlab = "Math Score",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")

boxplot(Science_Score ~ Gender, data = student,
        main = "Science Scores by Gender",
        xlab = "Gender",
        ylab = "Science Score",
        col = c("pink", "lightgreen"))
```

```
+   Math_Score = c(62, 80, 55, 90, 72, 82),
+   Science_Score = c(65, 85, 58, 92, 74, 86),
+   Exam_Date = as.Date(c("2025-01-10", "2025-01-10",
+                         "2025-02-12", "2025-02-12",
+                         "2025-03-15", "2025-03-15"))
+ )
>
> hist(student$Math_Score,
+      main = "Distribution of Math Scores",
+      xlab = "Math Score",
+      ylab = "Frequency",
+      col = "skyblue",
+      border = "black")
>
> boxplot(Science_Score ~ Gender, data = student,
+         main = "Science Scores by Gender",
+         xlab = "Gender",
+         ylab = "Science Score",
+         col = c("pink", "lightgreen"))
>
```

Environment	
student	6 obs. of 7 variables
survey_data	3 obs. of 4 variables
tdm	List of 6
Temp_mat	num [1:30, 1:30] 14.9 14.7 14.4 14.2 14 ...
top_days	3 obs. of 3 variables
traffic	3 obs. of 2 variables
traffic_data	3 obs. of 3 variables
ts_data	24 obs. of 4 variables
ui	List of 4



2.

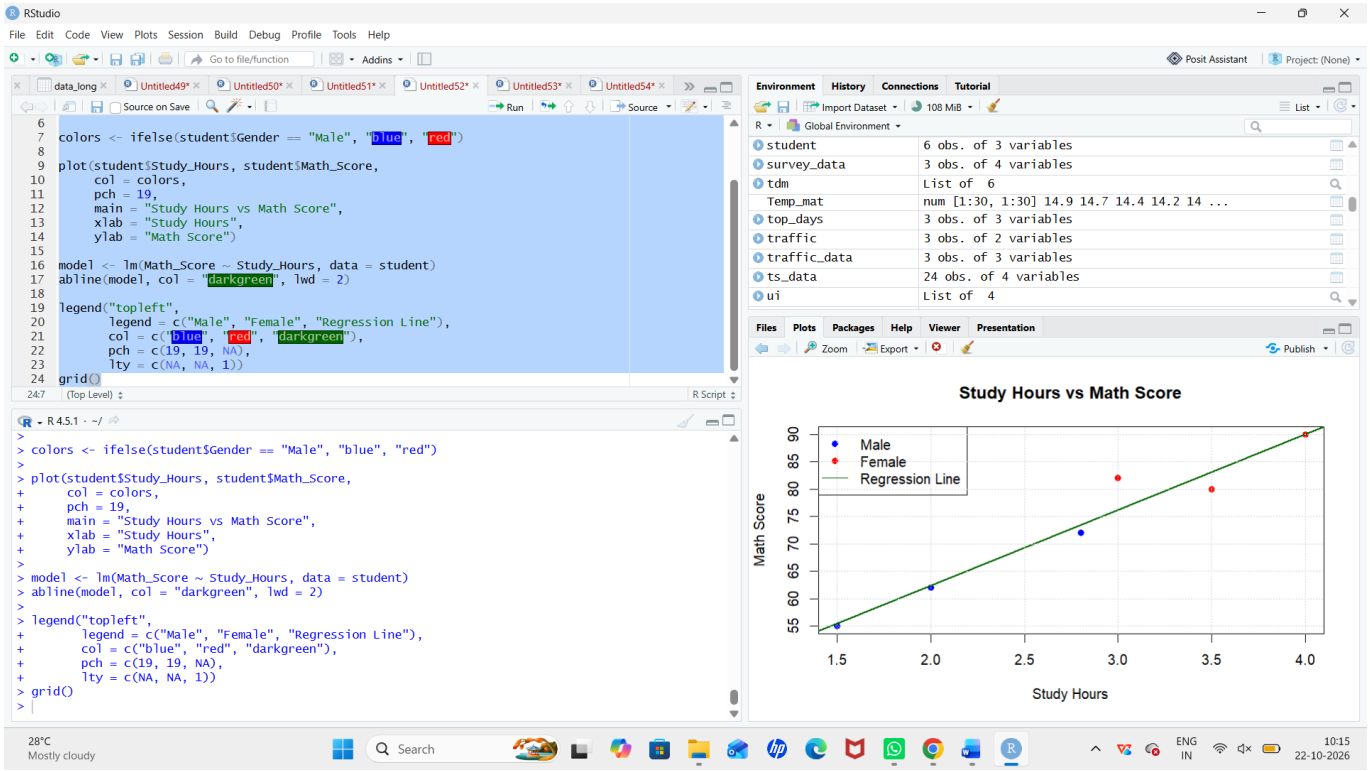
```
student <- data.frame(
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
  Math_Score = c(62, 80, 55, 90, 72, 82)
)

colors <- ifelse(student$Gender == "Male", "blue", "red")

plot(student$Study_Hours, student$Math_Score,
     col = colors,
     pch = 19,
     main = "Study Hours vs Math Score",
     xlab = "Study Hours",
     ylab = "Math Score")

model <- lm(Math_Score ~ Study_Hours, data = student)
abline(model, col = "darkgreen", lwd = 2)

legend("topleft",
       legend = c("Male", "Female", "Regression Line"),
       col = c("blue", "red", "darkgreen"),
       pch = c(19, 19, NA),
       lty = c(NA, NA, 1))
grid()
```



3.

```
student <- data.frame(
  Exam_Date = c("2025-01-10", "2025-01-10",
                "2025-02-12", "2025-02-12",
                "2025-03-15", "2025-03-15"),
  Math_Score = c(62, 80, 55, 90, 72, 82)
)

student$Exam_Date <- as.Date(student$Exam_Date)
student$Month <- format(student$Exam_Date, "%Y-%m")

monthly_avg <- aggregate(
  Math_Score ~ Month,
  data = student,
  FUN = mean
)

print(monthly_avg)

plot(monthly_avg$Math_Score,
     type = "o",
     pch = 19,
     col = "blue",
     xaxt = "n",
     main = "Monthly Average Math Score Trend",
     xlab = "Month",
     ylab = "Average Math Score")

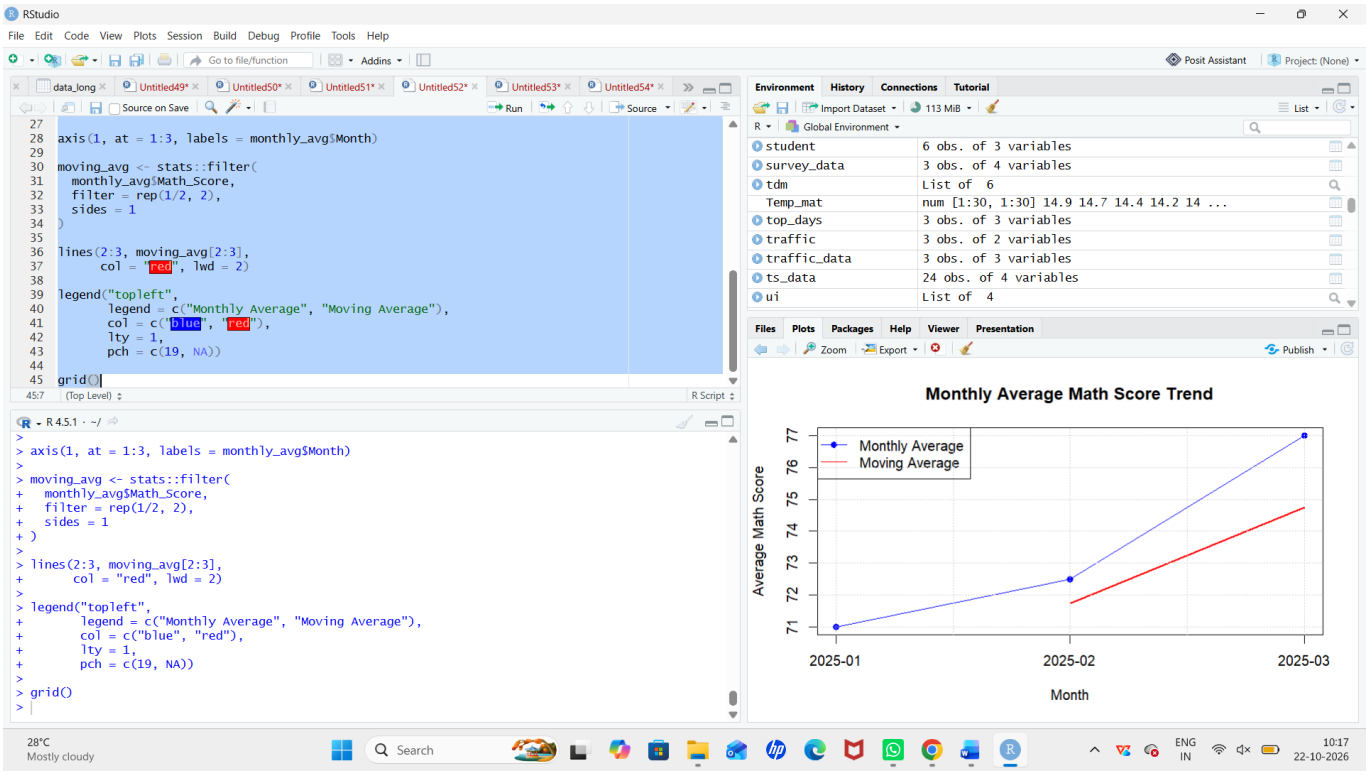
axis(1, at = 1:3, labels = monthly_avg$Month)

moving_avg <- stats::filter(
  monthly_avg$Math_Score,
  filter = rep(1/2, 2),
  sides = 1
)

lines(2:3, moving_avg[2:3],
      col = "red", lwd = 2)

legend("topleft",
       legend = c("Monthly Average", "Moving Average"),
       col = c("blue", "red"),
       lty = 1,
       pch = c(19, NA))

grid()
```



4.

- Save the dataset as Student_Mini_Data.csv.
- Open Tableau and connect to the CSV file.
- Check that Exam_Date is recognized as a Date field.
- Create Sheet 1 – Average Math Score by Gender:
  - Drag Gender to Columns.
  - Drag Math_Score to Rows.
  - Change the aggregation to Average.
  - Select a bar chart.
- Create Sheet 2 – Score Trend Over Time:
  - Drag Exam_Date to Columns.

- Drag Math_Score to Rows.
- Change the aggregation to Average and select a line chart.

 Add Exam_Date to Filters and choose the required date range.

 Create a new dashboard and add both worksheets.

 Display the date filter and add suitable chart titles and labels.

Procedure

1. Import the student dataset containing gender, study hours, attendance, Math scores, Science scores, and exam dates.
2. Create a histogram to visualize the distribution of Math scores.
3. Generate a boxplot to compare Science scores by gender.
4. Create a scatter plot showing the relationship between study hours and Math scores, using different colors for gender and adding a regression line.
5. Convert exam dates into Date format and calculate monthly average Math scores.
6. Plot the monthly score trend and apply moving average smoothing.
7. Import the dataset into Tableau and create a bar chart showing average Math scores by gender.
8. Create a line chart showing score trends over time and add an Exam_Date filter.
9. Combine the charts into an interactive dashboard and interpret the results.

Result : Student performance data was analyzed successfully using R and Tableau. The histogram and boxplot displayed score distributions, the scatter plot showed a positive relationship between study hours and Math scores, and the monthly trend chart showed a gradual increase in average Math scores. The Tableau dashboard presented gender-wise performance and score trends with an interactive date filter.